

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (09/06)

You have three international bank accounts: USD, EUR, JPY. You hire a foreign financial advisor to manage your accounts. But with your limited knowledge of the foreign language, you only know how to say one sentence:

“Please deduct from X account the current balance of Y account, and if the resulting balance is ≤ 0 , ignore my next Z sentences.”

- $[X] = [X] - [Y]$, then if $[X] \leq 0$, skip next Z;
- Shorthand: **please X, Y, Z**

You also have **empty** line of credit accounts (LOC1, LOC2, LOC3, LOC4, ...) for transfers.

1. Write a sequence of sentences to ask your financial advisor to transfer all funds in **both** your EUR and JPY accounts to your USD account.

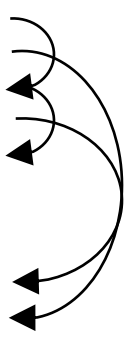
Answer:

```
please LOC1, EUR, 0
please LOC1, JPY, 0
please USD, LOC1, 0
please EUR, EUR, 0
please JPY, JPY, 0
please LOC1, LOC1, 0                # end (clean up, optional)
```

2. Write a sequence of sentences to ask your financial advisor to transfer all funds in both your EUR and JPY accounts to your USD account only if both accounts (EUR and JPY) currently have the **same** balance. Otherwise, keep the balances unchanged.

Answer:

```
please LOC1, EUR, 0
please LOC2, JPY, 0
please LOC1, LOC2, 1
please LOC4, LOC4, 7
please LOC3, LOC1, 1
please LOC4, LOC4, 4
please USD, LOC2, 0
please USD, LOC2, 0
please EUR, EUR, 0
please JPY, JPY, 0
please LOC3, LOC3, 0
please LOC2, LOC2, 0
please LOC1, LOC1, 0
```



```
# if ([JPY] - [EUR]) > 0...
#   ...then jump to end
# if -([JPY] - [EUR]) > 0...
#   ...then jump to end
# else, transfer funds to USD

# end (clean up, optional)
```